

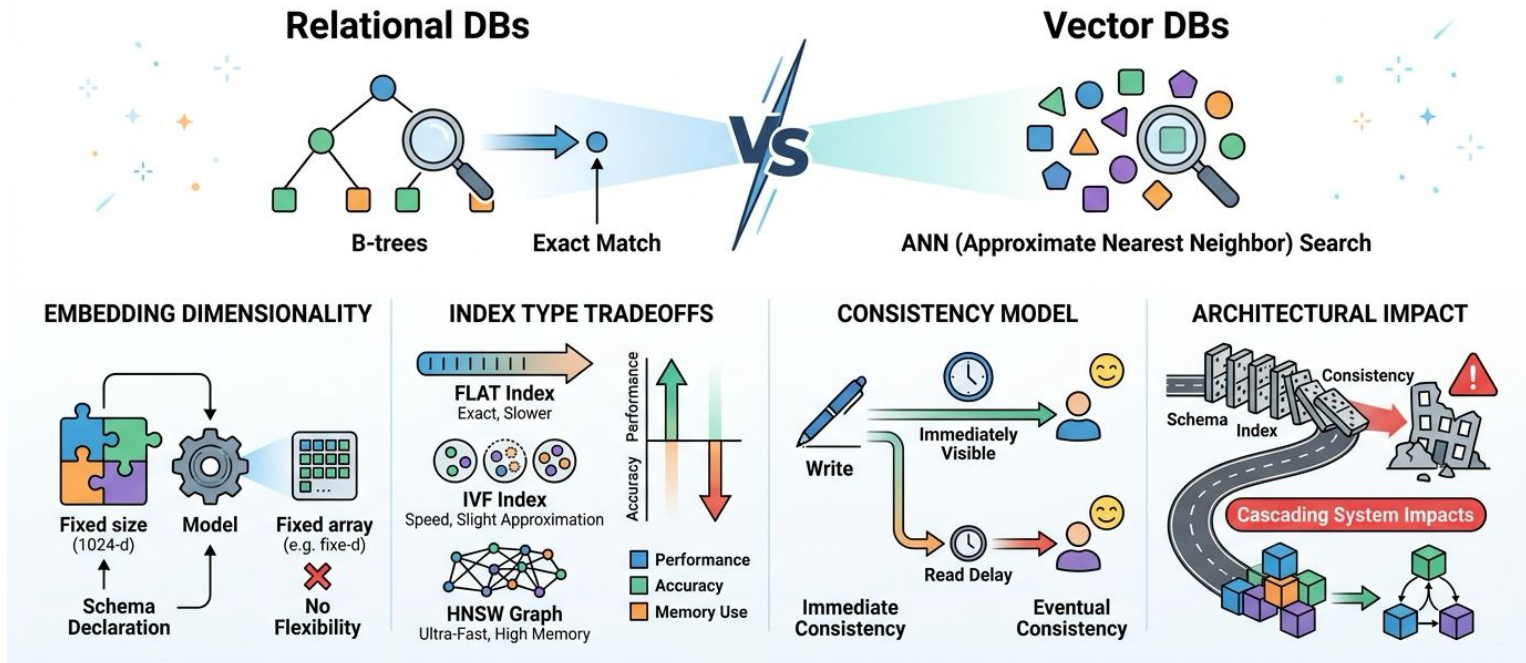


U.S. General Services
Administration

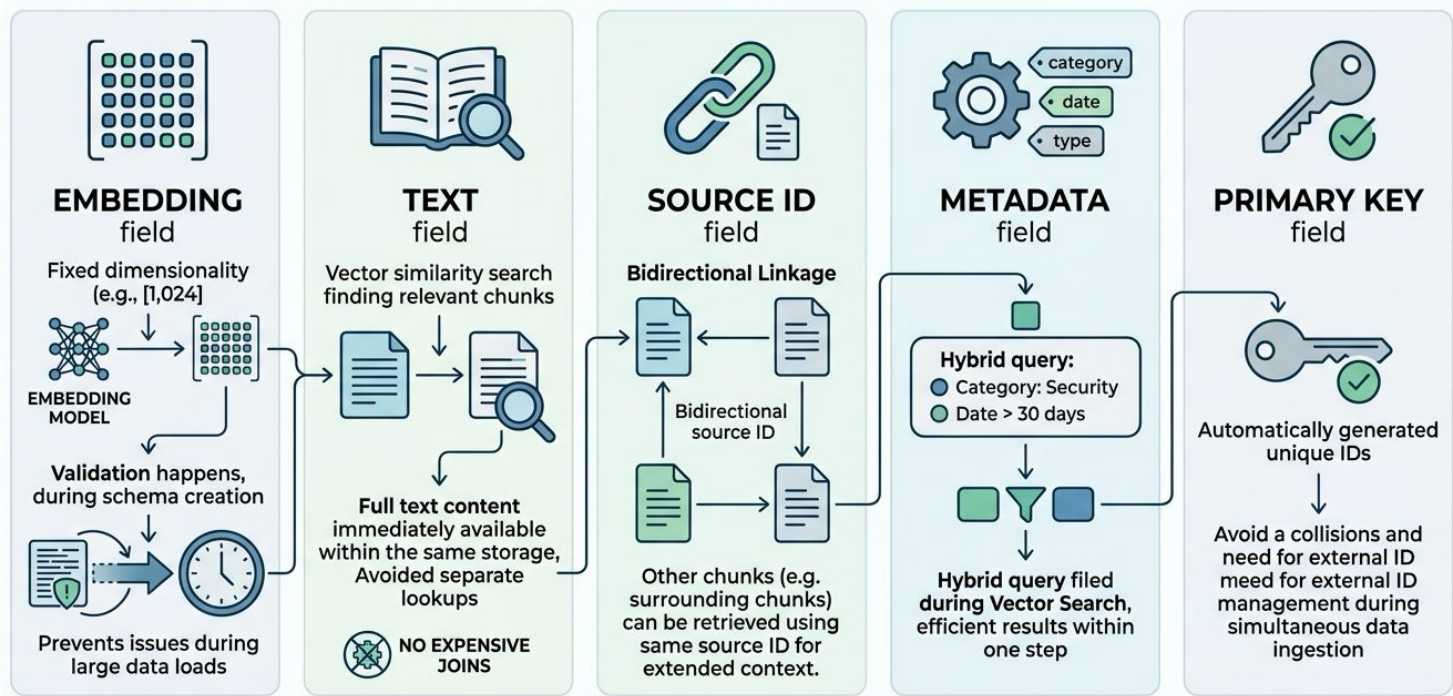
Chapter 6.3B: ETL Load Phase & Pipeline Integration

US General Service Administration
AI Community of Practice - March 2026

Vector DB Architecture Fundamentals

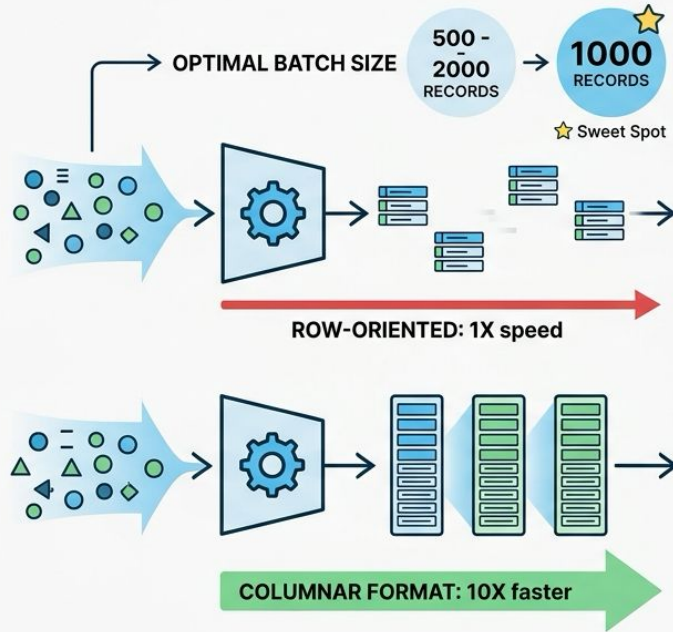


Collection Schema Design

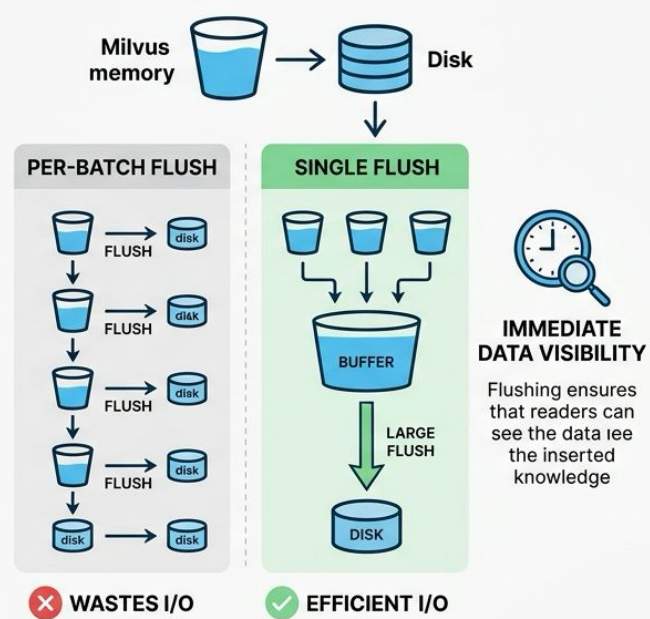


Batch Insertion Strategy

BATCHING STRATEGY

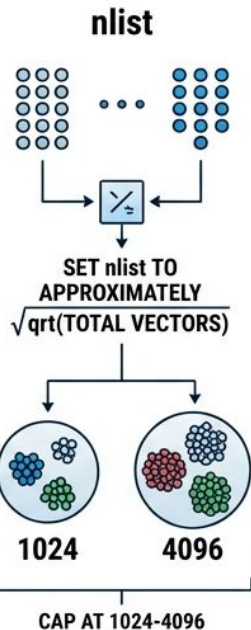
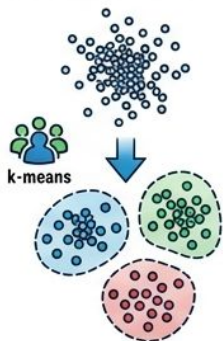


CONSISTENCY CONTROL



Vector Indexing for Millisecond Retrieval

IVF_FLAT
IVF_FLAT
PARTITIONS VECTORS INTO
CLUSTERS VIA k-MEANS



METRIC TYPE

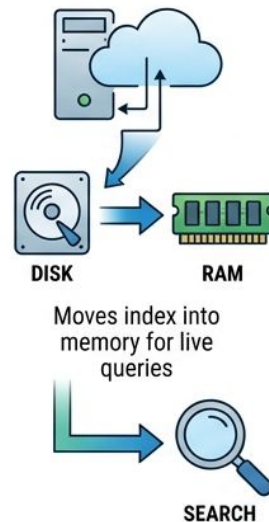


RETRIEVAL QUALITY
DEGRADES BY UP TO 40%

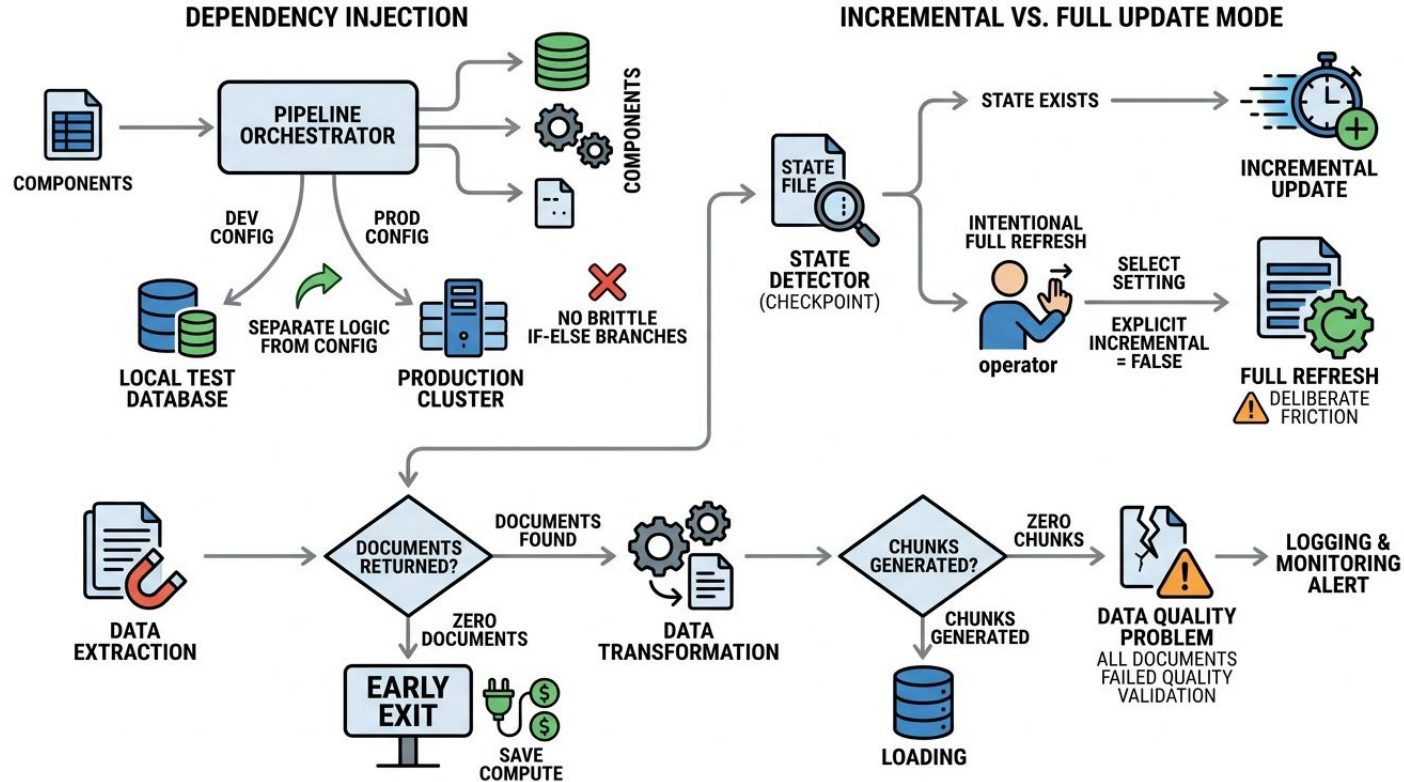
GOOD



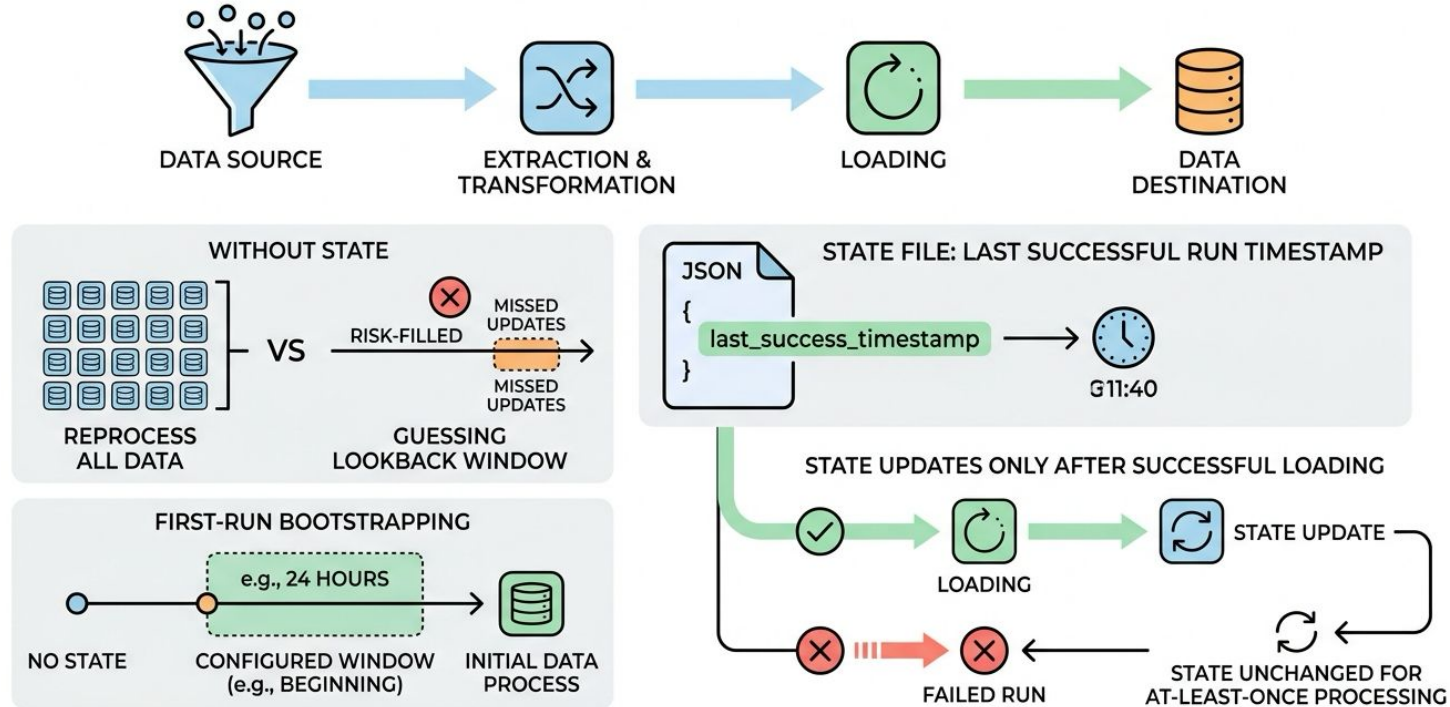
COLLECTION.LOAD()



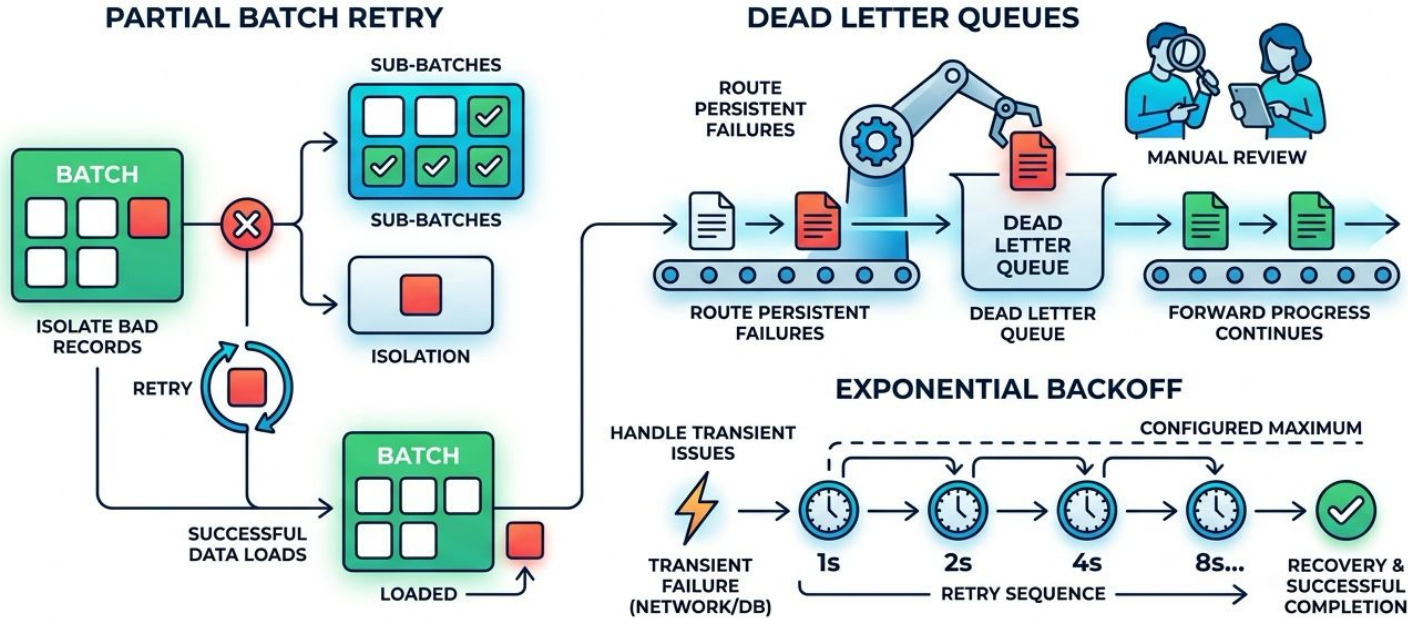
Pipeline Orchestration Architecture



State Management for Incremental Updates



Error Handling and Recovery Patterns



Monitoring and Observability

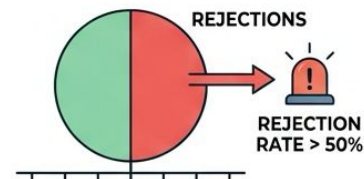
KEY METRICS TRACKING



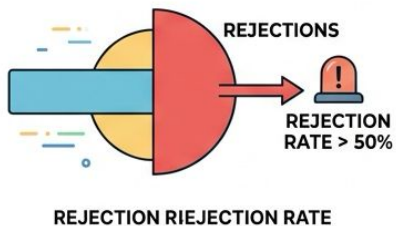
ALERT: ZERO EXTRACTIONS



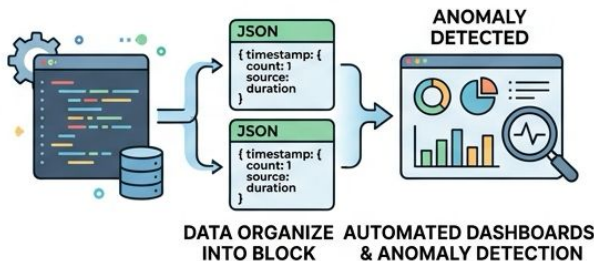
ALERT: HIGH REJECTION RATE



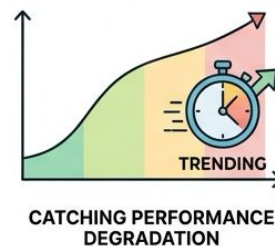
ALERT: HIGH REJECTION RATE



STRUCTURED LOGGING



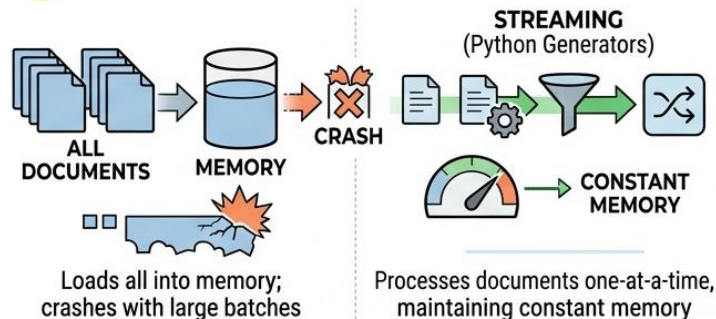
PERFORMANCE TRENDING



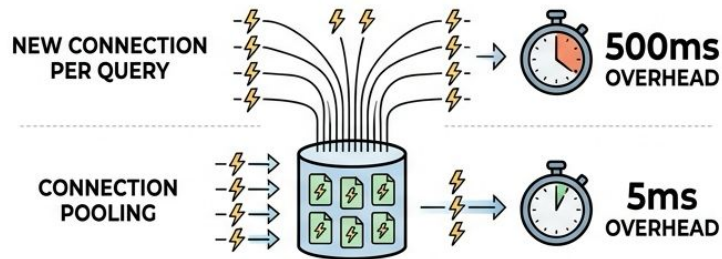
Streaming, Parallelism, and Connection Pooling



STREAMING TRANSFORMATIONS



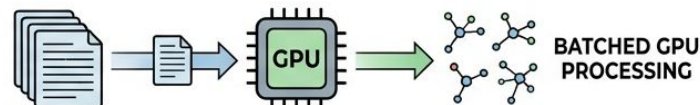
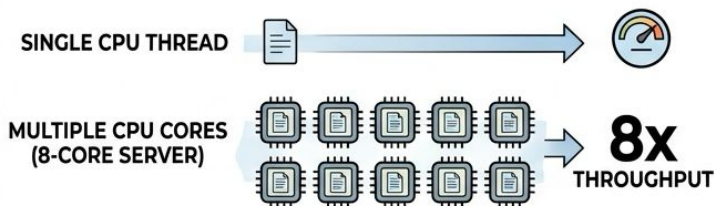
CONNECTION POOLING & EFFICIENCY



Reuses established connections; drops overhead from 500ms to 5ms

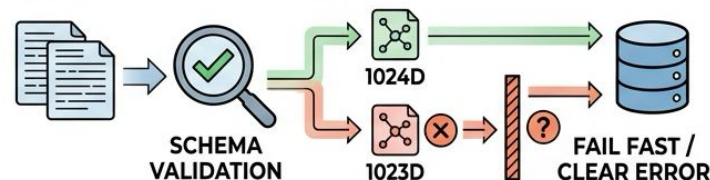


PARALLEL PROCESSING & EMBEDDING



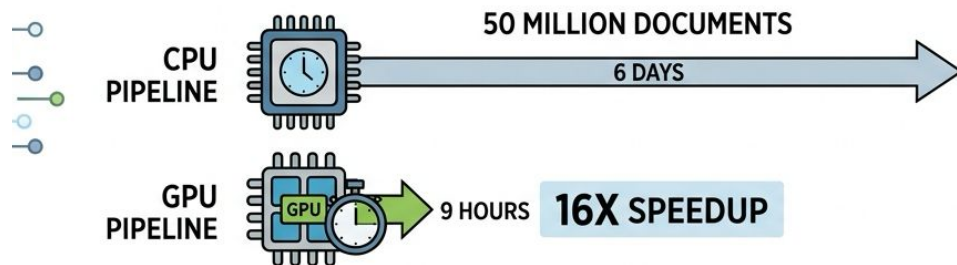
Leverages multiprocessing for parallelism and batching for GPU embeddings

SCHEMA VALIDATION

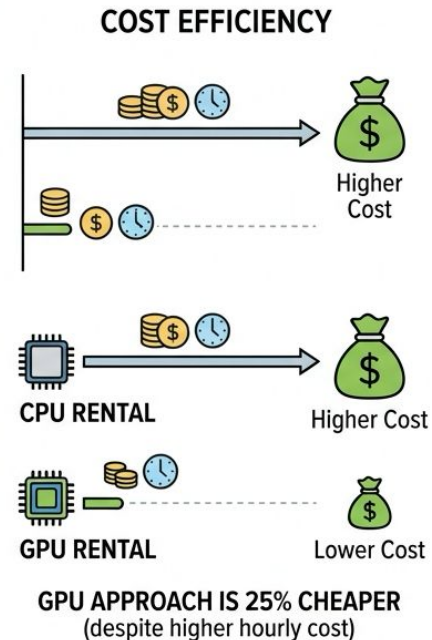
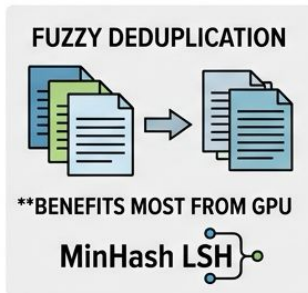
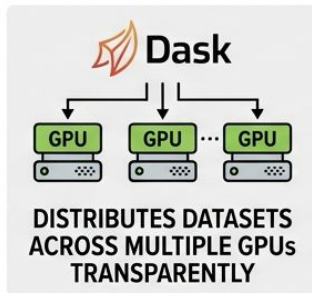
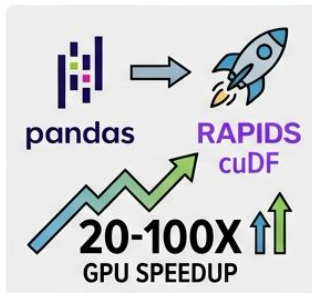


Catches malformed embeddings before database insertion

GPU-Accelerated ETL with NeMo Curator



KEY ENABLERS OF GPU PERFORMANCE



Key Patterns

